

DocuMind Cloud Architecture & Infrastructure Specification

1. System Overview & Architecture

- The platform is deployed across multiple cloud availability zones using Kubernetes (EKS/GKE).
- Core microservices communicate asynchronously via Apache Kafka message brokers.
- API Gateway manages rate limiting, authentication, and ingress routing.

2. Reliability & High Availability SLAs

- The infrastructure is designed for 99.99% annual uptime service level agreement (SLA).
- Recovery Time Objective (RTO) is strictly under 15 minutes in disaster recovery scenarios.
- Recovery Point Objective (RPO) is under 1 minute via synchronous cross-region transaction logs.

3. Performance & Latency Budgets

- P95 API response latency target is under 120 milliseconds for cached endpoints.
- P99 API response latency budget is capped at 250 milliseconds under peak concurrent load of 50,000 RPS.
- Vector database semantic search latency SLA is under 45 milliseconds for top-k retrieval.

4. Data Persistence & Security

- Primary relational data resides in PostgreSQL with multi-AZ replication and automated failover.
- Automated point-in-time recovery (PITR) backups are maintained for a 30-day retention window.
- All data is encrypted in transit using TLS 1.3 and at rest using AES-256 keys managed via KMS.